

HAT-P-36b

R-0913

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-36_230524 · created 2026-07-29T06:31:15+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2460088.787 +/- 0.0028 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1125 +/- 0.0095
Transit depth (Rp/Rs)^2	1.27 +/- 0.21 [%]
Orbital Inclination (inc)	86.36 +/- 1.91
Airmass coefficient 1 (a1)	1.157 +/- 0.013
Airmass coefficient 2 (a2)	-0.1175 +/- 0.0052
Scatter in the residuals of the lightcurve fit is	1.16 %
Best Comparison Star	None
Optimal Aperture	4.18
Optimal Annulus	13.54
Transit Duration (day)	0.0928 +/- 0.0038

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.1125 $\pm$ 0.0095 vs 0.126 (deviation 0.94 σ)
Duration fit vs published	0.0928 d vs 0.0925 d (deviation 0.05 σ)
Mid-time O–C	10.3 min
Depth SNR	7.8
Residual scatter	1.16 %

Light curve

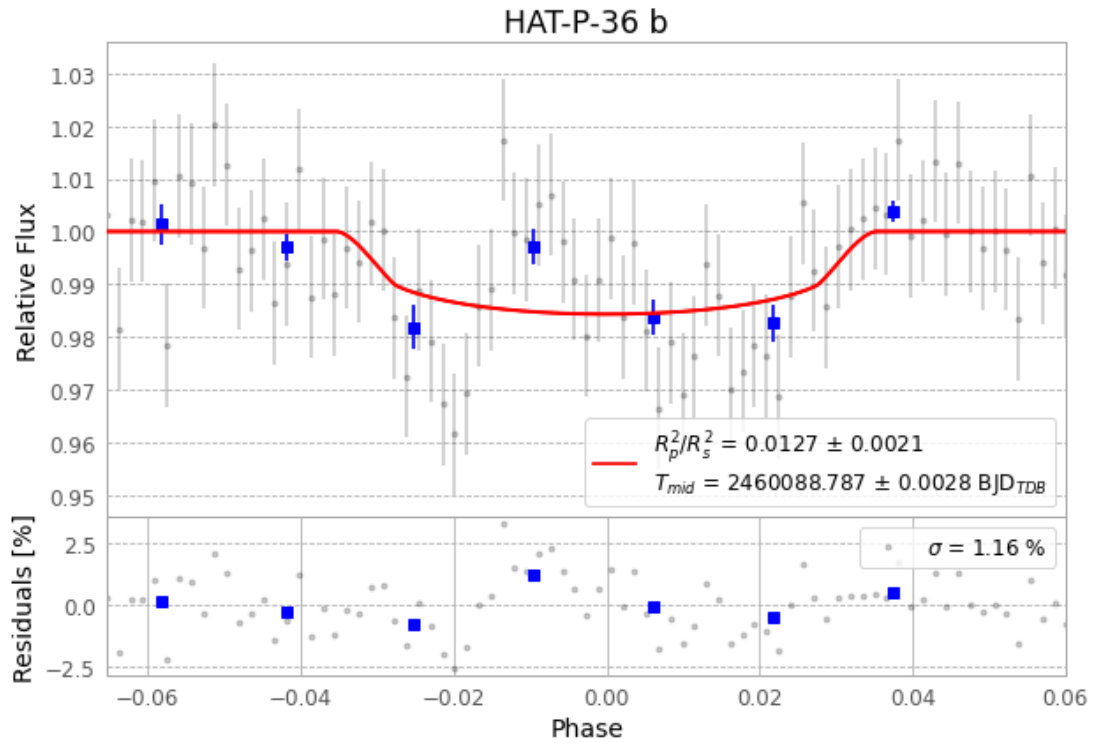


Figure 1 — FinalLightCurve_HAT-P-36 b_2023-05-23.png (SHA-256 a466d3bdc6429683...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [309, 347], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 8ae6dda47eda74c0...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

81 FITS frames (53 MB), HATP-36230524044515.FITS → HATP-36230524084515.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-36-230524.log (2f79724c88ca...), FinalLightCurve_HAT-P-36 b_2023-05-23.png (a466d3bdc642...), FinalLightCurve_HAT-P-36 b_2023-05-23.csv (3a0e80fedac9...)

Compute resources

Wall time 115.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-29 06:31Z.